

Manage Logs and Archive

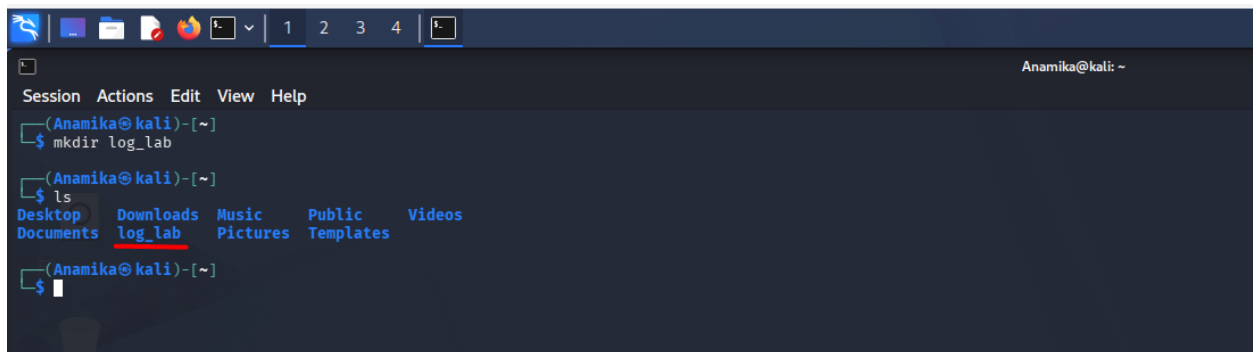
📁 Short Lab: Manage Logs and Archive

🎯 Goal

Create logs, archive them, delete them, and restore them.

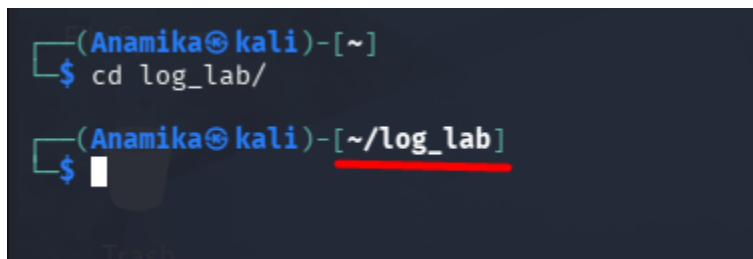
Step 1: Create log files

mkdir log_lab



A terminal window titled "Anamika@kali: ~" showing the execution of the command `mkdir log_lab`. The prompt is `(Anamika@kali)~$`. After running the command, the prompt changes to `(Anamika@kali)~$`. Below the prompt, the `ls` command is shown, and the output displays the directory structure: `Desktop Downloads Music Public Videos Documents log_lab Pictures Templates`. The `log_lab` directory is highlighted in red in the output.

cd log_lab



A terminal window titled "Anamika@kali: ~" showing the execution of the command `cd log_lab/`. The prompt is `(Anamika@kali)~$`. After running the command, the prompt changes to `(Anamika@kali)~/log_lab$`. The `~/log_lab` part of the prompt is highlighted in red.

touch app.log error.log access.log

Verify:

ls

```
(Anamika@kali)-[~/log_lab]
$ touch app.log error.log access.log

(Anamika@kali)-[~/log_lab]
$ ls
access.log  app.log  error.log

(Anamika@kali)-[~/log_lab]
$
```

Step 2: Add sample logs

echo "Application Started" > app.log

echo "Database Error" > error.log

echo "User Login" > access.log

```
(Anamika@kali)-[~/log_lab]
$ echo "Application started" > app.log

(Anamika@kali)-[~/log_lab]
$ echo "Database error" > error.log

(Anamika@kali)-[~/log_lab]
$ echo "user login" > access.log

(Anamika@kali)-[~/log_lab]
$
```

Step 3: Archive logs

tar-czvf logs_backup.tar.gz *.log

Meaning

- c = Create
- z = Compress (gzip)
- v = Show files

- f = Archive file name

```
(Anamika@kali)-[~/log_lab]
$ tar -czvf logs_backup.tar.gz *.log
access.log
app.log
error.log

(Anamika@kali)-[~/log_lab]
$
```

Step 4: Verify archive contents

tar-tzvf logs_backup.tar.gz

Expected:

app.log

error.log

access.log

```
(Anamika@kali)-[~/log_lab]
$ tar -tzvf logs_backup.tar.gz
-rw-rw-r-- Anamika/Anamika  11 2026-06-07 23:07 access.log
-rw-rw-r-- Anamika/Anamika  20 2026-06-07 23:06 app.log
-rw-rw-r-- Anamika/Anamika  15 2026-06-07 23:07 error.log

(Anamika@kali)-[~/log_lab]
$
```

Step 5: Simulate log loss

rm *.log

Verify:

ls

Output:

logs_backup.tar.gz

```
(Anamika@kali)-[~/log_lab]
$ rm *.log

(Anamika@kali)-[~/log_lab]
$ ls
logs_backup.tar.gz

(Anamika@kali)-[~/log_lab]
$
```

Step 6: Restore logs

tar-xzvf logs_backup.tar.gz

```
(Anamika@kali)-[~/log_lab]
$ tar -xzvf logs_backup.tar.gz
access.log
app.log
error.log

(Anamika@kali)-[~/log_lab]
$
```

Step 7: Verify restoration

ls

Output:

access.log

app.log

error.log

logs_backup.tar.gz

```
(Anamika@kali)-[~/log_lab]
$ ls
access.log  app.log  error.log  logs_backup.tar.gz

(Anamika@kali)-[~/log_lab]
$
```

Interview Explanation

"I practiced log management by creating log files, archiving them using tar and gzip, verifying archive contents, simulating data loss, and restoring logs from backups."

Skills Practiced

- ☒ Log Management
- ☒ Backup & Restore
- ☒ Archive Creation
- ☒ Disaster Recovery Basics
- ☒ Linux Administration